

2023 AMC 10A Problem 8 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 8. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 8

Problem:

Barb the baker has developed a new temperature scale for her bakery called the Breadus scale, which is a linear function of the Fahrenheit scale. Bread rises at 110 degrees Fahrenheit, which is 0 degrees on the Breadus scale. Bread is baked at 350 degrees Fahrenheit, which is 100 degrees on the Breadus scale. Bread is done when its internal temperature is 200 degrees Fahrenheit. What is this in degrees on the Breadus scale?

Answer Choices:

- A. 33
- B. 34.5
- C. 36
- D. 37.5
- E. 39

Solution:

The difference between the bread rising temperature and the bread baking temperature in degrees Fahrenheit is $350 - 110 = 240$. The Fahrenheit temperature of bread when it is done baking is 200 degrees, which is $200 - 110 = 90$ degrees above that of bread rising, so is $\frac{90}{240} = \frac{3}{8}$ of the way from 110 to 350. Therefore on the Breadus scale this is $\frac{3}{8}$ of the way from 0 to 100. This value is

$$\frac{3}{8} \cdot (100 - 0) = \frac{300}{8} = (D) \boxed{37.5}$$

OR

The graph of the number of degrees Breadus (y -axis) as a function of the number of degrees Fahrenheit (x -axis) is a line that passes through the points $(110, 0)$ and $(350, 100)$. The slope of this line is

$$\frac{100 - 0}{350 - 110} = \frac{5}{12}$$

The equation of the line is therefore $y = \frac{5}{12}(x - 110)$, and the value of y when $x = 200$ is $\frac{5}{12} \cdot 90 = \text{(D)} \boxed{37.5}$.

OR

The internal temperature of bread in degrees Fahrenheit when it is done baking is a weighted average (a convex combination) of the extremes of 110 and 350, so $110x + 350(1 - x) = 200$ for some value x . Solving for x gives $x = \frac{5}{8}$. Substituting this in the convex combination for internal temperature of bread in degrees Breadus when it is done baking gives

$$0x + 100(1 - x) = 0 \cdot \frac{5}{8} + 100 \cdot \frac{3}{8} = \text{(D)} \boxed{37.5}$$

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